Junyao Dong

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Education

Nanyang Technological University — Singapore
M.Eng. by Research in Computing and Data Science Engineering — 08/2024 – 06/2026
- GPA: 4.17/5.0. Co-advised by Cheng Long and Shuhao Zhang.
- **Research Interests:** Vector databases, retrieval systems, and machine learning systems.

The University of Manchester — Manchester, UK
Visiting Student in Computer Science — 08/2019 – 05/2020
- Overall Average: 70% (First-Class Standard).

Guangdong University of Foreign Studies — Guangzhou, China
B.Eng. in Computer Science — 09/2017 – 06/2021
- GPA: 3.4/4.0.

Publications (* = Equal contribution)

The Program Counter Is a Path: Code, Data, and Control in One Addressable Tree
Peng Li*, Junyao Dong*
- Submitted to *Proceedings of the ACM on Programming Languages*, 2027 (**PLDI'27**). [Website] [Code]

Shared Memory in Multi-Agent Systems: Which View Does the Agent Read?
Junyao Dong, Xisen Wang, Yu Chen, Yi Li, Qi Zhang
- Submitted to the *26th International Conference on Autonomous Agents and Multiagent Systems* (**AAMAS'27**).

ANNchor: Concurrency Control for Freshness in Streaming Graph-Based Approximate Nearest Neighbor Search
Junyao Dong, Shuhao Zhang, Wenchao Zhou, Cheng Long
- Revision at *Proceedings of the ACM on Management of Data*, 2027 (**SIGMOD'27**, Round 3; decision Dec 12)[1]. [Latest Status]

CANDOR-Bench: Benchmarking In-Memory Continuous ANNS under Dynamic Open-World Streams
Mingqi Wang*, Junyao Dong*, Zhuoyan Wu, Jun Liu, Ruicheng Zhang, Jianjun Zhao, Ruipeng Wan, Xinyan Lei, Shuhao Zhang, Bolong Zheng, Haikun Liu, Xiaofei Liao, Hai Jin
- *Proceedings of the ACM on Management of Data*, 2026 (**SIGMOD'26**). [Paper] [Code]

Research Experience († = Full-time)

Vector Database Group @ NTU — Singapore
Research Associate† (Advised by Cheng Long) — 08/2026 – Present

Applying vector search and RaBitQ quantization to long-context LLM inference.
- Studying retrieval over the KV cache for sparse attention in long-context decoding.

PageIndex — Remote
Research Intern (Co-advised by Mingtian Zhang and Yu Tang) — 12/2025 – Present

Building retrieval over long, hierarchically structured documents.
- Led the design and implementation of ConDB, a chunk-free, vector-free context database for long documents.
- Collaborated with MongoDB on storing and searching large hierarchical JSON in the MongoDB Python driver.

Vector Database Group @ NTU & IntelliStream @ HUST — Singapore
Graduate Student Researcher† (Co-advised by Cheng Long and Shuhao Zhang) — 08/2024 – 06/2026

Investigated index freshness in streaming vector search, and how to benchmark it.
- Designed ANNchor, a concurrency-control framework for post-acknowledgment visibility on one live HNSW graph; up to 4.6× search throughput over global locking and 68% lower P99 latency under write-heavy load.
- Co-developed CANDOR-Bench, a continuous ANNS benchmark of 19 algorithms on 12 datasets under drift and concurrent read–write streams; co-designed the streaming evaluation harness.

[1] SIGMOD reviews papers in rolling rounds. For Round 3, the first notification (accept, reject, or revise) was on Oct 19, 2026; the revised paper is due Nov 19, and the final accept/reject decision is announced on Dec 12, 2026. The latest status will be posted on my homepage.

The Hong Kong University of Science and Technology (Guangzhou) Guangzhou, China
Research Assistant† (Advised by Ying Cui) 04/2024 – 06/2024

Implemented adaptive video delivery for variable network conditions.

- Developed an adaptive bitrate system that cuts playback buffering without losing visual quality.

Selected Open-Source Projects

RaBitQ-Library ∼300 GitHub stars
Contributor github.com/VectorDB-NTU/RaBitQ-Library

Official library for RaBitQ, a randomized bitwise quantization method for approximate nearest neighbor search, covering quantization, scanning, and index integration (used by Meta, NVIDIA, Microsoft, and Alibaba).

- Hardened IVF index construction robustness and prototyped quantized graph-index search integration.

RL-Kernel 400+ GitHub stars
Contributor github.com/RL-Align/RL-Kernel

High-performance RL post-training infrastructure for operator-level train–inference consistency.

- Implemented CUDA/Triton kernels for DeepSeek V4 compressed attention: one-softmax MQA with an attention sink, and grouped output projection.

vllm-rlt 200+ GitHub stars
Contributor github.com/InferMatrixLab/vllm-rlt

Inference engine for looped language models. Recurrent steps are batched continuously, so requests at different loop depths share one batch.

- Designed the engine loop so synchronous and asynchronous steps share scheduling and result handling, while submission, GPU completion, and output delivery stay separate.

Industry Experience († = Full-time)

Baidu Shanghai, China
Software Engineer†, Cloud Services 05/2021 – 03/2024

Built live-video infrastructure, from edge ingest through storage to on-air production.

- Integrated H.264/H.265 at the edge for Baidu Cloud Live Streaming Service and wrote outputs as MP4, FLV, and HLS.
- Added watermarks, picture-in-picture, transitions, and subtitles to Baidu Cloud Live Cast Production Studio on a customized FFmpeg engine and Baidu's RTC stack.

Momenta Suzhou, China
Research and Development Intern, Autonomous Driving 10/2020 – 05/2021

Developed log infrastructure for replaying recorded drives as training data.

- Synchronized perception and planning streams onto one timeline.

Awards and Honors

Second Prize, National Modeling Competition 2020

Overseas Student Programme Award 2020

Academic Service

Availability and Reproducibility Committee (ARC)

- SIGMOD 2026 2026

External Reviewer

- VLDB 2026, SIGMOD 2027, VLDB 2027, WSDM 2027 2026
- ICDE 2026 2025